

1. Master equation for the stochastic SIRI model

State variables

At any time t , the system is described by the state

$$(S, I, R), \quad \text{with } S + I + R = N.$$

Let



$$P(S, I, R; t)$$

be the probability that the system is in state (S, I, R) at time t .

Reactions and transition rates

Reaction	State change	Transition rate
Infection: $S + I \rightarrow I + I$	$(S, I, R) \rightarrow (S - 1, I + 1, R)$	$\frac{\alpha SI}{N}$
Recovery: $I \rightarrow R$	$(S, I, R) \rightarrow (S, I - 1, R + 1)$	βI
Relapse: $R \rightarrow I$	$(S, I, R) \rightarrow (S, I + 1, R - 1)$	γR

Master equation

The Master equation balances **gain** and **loss** of probability for each state:

$$\begin{aligned} \frac{d}{dt}P(S, I, R; t) = & \underbrace{\frac{\alpha(S + 1)(I - 1)}{N}P(S + 1, I - 1, R; t)}_{\text{gain from infection}} \\ & + \underbrace{\beta(I + 1)P(S, I + 1, R - 1; t)}_{\text{gain from recovery}} \\ & + \underbrace{\gamma(R + 1)P(S, I - 1, R + 1; t)}_{\text{gain from relapse}} \\ & - \underbrace{\left[\frac{\alpha SI}{N} + \beta I + \gamma R\right]P(S, I, R; t)}_{\text{loss from current state}}. \end{aligned}$$

Physical meaning of the terms

- **Gain terms:** Probability flows *into* the state (S, I, R) from neighboring states due to one reaction occurring.
 - **Loss term:** Probability flows *out of* the state (S, I, R) because any of the three reactions can occur from it.
 - The factor SI/N reflects **mass-action mixing**: infection requires contact between susceptible and infected individuals.
 - The recovery and relapse terms are linear because they involve single individuals transitioning independently.
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2. Gillespie algorithm pseudocode for stochastic simulation

The Gillespie algorithm simulates the **exact stochastic dynamics** implied by the Master equation.

Pseudocode

```
Set time  $t = 0$ 
Set final simulation time  $T$ 

While  $t < T$  and  $I > 0$ :
  Compute propensities:
     $a_1 = \alpha * S * I / N$       # infection
     $a_2 = \beta * I$               # recovery
     $a_3 = \gamma * R$             # relapse

  Compute total rate:
     $a_0 = a_1 + a_2 + a_3$ 

  Draw two random numbers  $r_1, r_2 \sim \text{Uniform}(0,1)$ 

  Compute time to next reaction:
     $\Delta t = (1 / a_0) * \ln(1 / r_1)$ 
  Update time:
     $t = t + \Delta t$ 

  Determine which reaction occurs:
    If  $r_2 * a_0 < a_1$ :
       $S = S - 1$ 
       $I = I + 1$ 
    Else if  $r_2 * a_0 < a_1 + a_2$ :
       $I = I - 1$ 
```



```

        R = R + 1
Else:
        R = R - 1
        I = I + 1

```

```

Record (t, S, I, R)
End While

```

Interpretation

- Each reaction is treated as a **random event** occurring with a rate determined by the current state.
- Time advances in random jumps, not fixed steps.
- This algorithm exactly reproduces the Master equation in the limit of many realizations.

3. Deterministic ODEs for the SIR model

When population sizes are large, stochastic fluctuations average out and the **mean-field equations** describe the dynamics.

ODE system

$$\begin{aligned}
 \frac{dS}{dt} &= -\frac{\alpha SI}{N}, \\
 \frac{dI}{dt} &= \frac{\alpha SI}{N} - \beta I + \gamma R, \\
 \frac{dR}{dt} &= \beta I - \gamma R.
 \end{aligned}$$



Physical meaning of each term

Susceptible equation

$$\frac{dS}{dt} = -\frac{\alpha SI}{N}$$

- Susceptibles decrease due to infection after contact with infected individuals.

Infected equation

$$\frac{dI}{dt} = \frac{\alpha SI}{N} - \beta I + \gamma R$$

- Gain from infection
- Loss from recovery
- Gain from relapse (loss of immunity)

Recovered equation

$$\frac{dR}{dt} = \beta I - \gamma R$$

- Recovered individuals increase due to recovery.
 - They decrease due to relapse back to infection.
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Question 2

1.

Let:

n_1 = number of steps to the right

$N - n_1$ = number of steps to the left

Each step is an independent Bernoulli trial with:

$$P(\text{right}) = 1/2$$

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The probability of taking exactly n_1 right steps in N trials is:

$$P(n_1) = {}^N C_{n_1} (1/2)^{n_1} (1/2)^{N-n_1} = {}^N C_{n_1} (1/2)^N$$

This is a binomial distribution with parameters N and $p = 1/2$.

MARKING : +2 if showed that it follows binomial distribution
0 otherwise

2.

Expected value of n_1

For a binomial random variable:

$$E[n_1] = Np$$

Since $p = 1/2$,

$$E[n_1] = N/2$$

(No proof required, as stated.)

MARKING : +1 if the final answer is correct
0 otherwise (no partial marking)

3.

Average displacement after N steps

Displacement formula

Each right step contributes $+d_0$, each left step contributes $-d_0$.

Total displacement:

$$x = d_0 (n_1 - (N - n_1)) = d_0 (2n_1 - N)$$

Average displacement

$$\langle x \rangle = d_0 (2\langle n_1 \rangle - N)$$

$$\begin{aligned}\text{Substitute } \langle n_1 \rangle &= N/2: \\ \langle x \rangle &= d_0 (N - N) = 0 \\ \langle x \rangle &= 0\end{aligned}$$

The walk is unbiased, so on average the walker stays at the origin.

MARKING : +2 if showed avg displacement = 0
+1 if the steps are partially correct, but final answer is wrong
0 otherwise

4.

Pseudocode to compute the mean displacement

```
Initialize: (0.5 marks)
    Set number of steps N
    Set step size  $d_0$ 
    Set number of realizations M
    Set sum_displacement = 0

For i = 1 to M: (1 mark)
    Set position x = 0
    For j = 1 to N: (1 mark)
        Draw random number r from Uniform(0,1)
        If r < 0.5:
            x = x +  $d_0$  # move right
        Else:
            x = x -  $d_0$  # move left
    End For
    sum_displacement = sum_displacement + x
End For
mean_displacement = sum_displacement / M
Output mean_displacement
```

As $M \rightarrow \infty$, the computed mean displacement approaches zero, matching the analytical result.

MARKING : +3 if the entire pseudocode is correct (with initialisation)
+2.5 if variables not initialised, but correct pseudocode
+1.5 if the outer for loop is missing
0 if pseudocode is wrong

Question 3: The Lotka-Volterra System

The Lotka-Volterra system is

$$\dot{x} = \alpha x - \beta xy, \quad \dot{y} = \delta xy - \gamma y$$

where $x(t)$ represents prey, $y(t)$ represents predators, and all parameters are positive.

1. Fixed Points

Fixed points satisfy $\dot{x} = 0$ and $\dot{y} = 0$.

From

$$\dot{x} = x(\alpha - \beta y) = 0$$

$$x = 0 \quad \text{or} \quad y = \frac{\alpha}{\beta}$$

From

$$\dot{y} = y(\delta x - \gamma) = 0$$

$$y = 0 \quad \text{or} \quad x = \frac{\gamma}{\delta}$$

Combining these results, the system has two equilibrium points

1. Extinction equilibrium

$$(x^*, y^*) = (0, 0)$$

2. Coexistence equilibrium

$$(x^*, y^*) = \left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta} \right)$$

2. Stability of the Fixed Points

We analyze stability using the Jacobian matrix

$$J(x, y) = \begin{pmatrix} \alpha - \beta y & -\beta x \\ \delta y & \delta x - \gamma \end{pmatrix}$$

(a) **Fixed Point** $(0, 0)$

$$J(0, 0) = \begin{pmatrix} \alpha & 0 \\ 0 & -\gamma \end{pmatrix}$$

Eigenvalues

$$\lambda_1 = \alpha > 0, \quad \lambda_2 = -\gamma < 0$$

Since the eigenvalues have opposite signs, this equilibrium is a saddle point and therefore unstable.

Biologically, prey grow exponentially when predators are absent, while predators die out without prey.

(b) Fixed Point $\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$

Evaluating the Jacobian

$$J = \begin{pmatrix} 0 & -\frac{\beta\gamma}{\delta} \\ \frac{\delta\alpha}{\beta} & 0 \end{pmatrix}$$

Eigenvalues satisfy

$$\lambda^2 = -\alpha\gamma$$

$$\lambda = \pm i\sqrt{\alpha\gamma}$$

The eigenvalues are purely imaginary, so the linearized system predicts a center and the equilibrium is neutrally stable.

The Lotka-Volterra equations admit a conserved quantity, so trajectories form closed orbits around this equilibrium.

Rubric for Question 3 (Total: 2 Marks)

Identification of Equilibrium States (0.5 marks)

- 0.25 marks for each correctly identified equilibrium state.
- 0 marks if an equilibrium state is missing or wrong.
- If four fixed points are written, deduct 0.25 marks from this part.

Eigenvalue Analysis and Stability Classification (1 mark)

- 0.5 marks for correct eigenvalues.
- 0.5 marks for correct stability classification of both equilibrium points.
- Deduct 0.25 marks for each case wherein the eigenvalues are correct but interpretation is wrong.
- Award 0 if eigenvalues are wrong even if the Jacobian is correct.

Long-Term Population Behaviour (0.5 marks)

- 0.25 marks for correct qualitative explanation for the long term population for each equilibrium.
- Award 0 in each case if the qualitative explanation is wrong.

Question 4: Randomness and Periodicity (2 Marks)

How would you determine whether a given **sequence** of data is randomly drawn from a uniform distribution and does not exhibit periodicity?

Marking Rubric

To verify the quality of a random sequence, two independent properties must be tested.

1. Uniformity (Distribution Test)

- Method: Plot a histogram of the data.
- Requirement: The histogram should be approximately flat across the interval, indicating equal probability for all values.
- Statistical alternative: Use a Chi-Squared (χ^2) test or a Kolmogorov-Smirnov (K-S) test to compare the empirical distribution to the theoretical uniform distribution.

2. Periodicity and Independence (Correlation Test)

- Method A (time/frequency domain): Examine the autocorrelation function (ACF) or Fourier spectrum (FFT). For random data, the ACF should be close to zero for lags $k > 0$ and the FFT should not contain strong periodic peaks.
- Method B (phase space): Construct a lag plot of x_n versus x_{n+1} . A random sequence produces an unstructured cloud, whereas periodic sequences produce visible geometric patterns.

Note on the Insufficiency of Moment-Based and Frequency-Based Methods

To determine if a sequence is truly randomly drawn from a Uniform distribution, standard statistical tests (like Chi-Squared or Kolmogorov-Smirnov) and correlation tests (like ACF or Lag Plots) must be used. Simpler heuristics often fail for the following reasons:

- **Moments are Non-Unique:** Matching the first few moments is a necessary but not sufficient condition.
 - A discrete distribution taking values $0.5 \pm \frac{1}{\sqrt{12}}$ with equal probability matches the mean ($\mu = 0.5$) and variance ($\sigma^2 = \frac{1}{12}$) of $U(0, 1)$ but is not uniform.
 - A distribution taking values $\{0, 0.5, 1\}$ with probabilities $\{1/6, 4/6, 1/6\}$ perfectly matches the first three raw moments ($\frac{1}{k+1}$ for $k = 1, 2, 3$) of $U(0, 1)$ despite being highly non-uniform.
- **Convergence $\neq$ Randomness:** The Strong Law of Large Numbers (SLLN) ensures the sample mean converges to the expected value $\frac{a+b}{2}$. However, a constant sequence (e.g., $x_i = 0.5$ for all i) satisfies this convergence but has zero variance and zero randomness.
- **Histograms and CDFs Destroy Order:** Periodicity is a temporal property. Tools that measure frequency (Histograms, CDFs) or sorting the data completely erase the sequence order.
 - A periodic sequence (1, 2, 3, 1, 2, 3) and a random shuffle of the same values (3, 1, 2, 2, 3, 1) will produce *identical* histograms and CDFs. Therefore, these tools cannot detect or disprove periodicity.
- **Sorting Destroys Patterns:** Sorting a sequence to check for uniformity is a "one-way street." Since any periodic sequence and any random sequence containing the same values will result in the same sorted list, sorting renders the detection of patterns mathematically impossible.

Do note that full marks **have not** been awarded if your answer is based on any of the above four methods.

Marking Deductions

Item	Marks	Deduction Criteria
Uniformity	1.0	Deduct 0.5 if a histogram is mentioned but the expected flatness is not explained. Deduct 1.0 if the answer ignores distribution testing entirely.
Periodicity	1.0	Deduct 0.5 if tools such as FFT or ACF are listed without describing what indicates randomness. Deduct 0.5 if a scatter plot is suggested but not specified as a lag plot (x_n vs x_{n+1}).
General		Deduct 0.5 if the student confuses distribution and periodicity. If only uniformity is discussed with no periodicity test, award 0. Example: the sequence $0.1, 0.2, 0.3, \dots, 1.0, 0.1, 0.2, \dots$ has a perfectly flat histogram but is completely periodic.

Question 5: Answer Key and Marking Scheme

Part 1: Normalization Constant and Normalized PDF

Given

$$f(x) = \sin x \cos x \quad 0 \leq x \leq \frac{\pi}{2}$$

A probability density must satisfy

$$\int_{-\infty}^{\infty} p(x) dx = 1$$

Let

$$p(x) = C \sin x \cos x$$

Compute the normalization constant

$$\int_0^{\pi/2} C \sin x \cos x dx = 1$$

Using $\sin(2x) = 2 \sin x \cos x$

$$C \frac{1}{2} \int_0^{\pi/2} \sin(2x) dx = C \frac{1}{2}$$

$$C = 2$$

Normalized PDF

$$p(x) = 2 \sin x \cos x = \sin(2x)$$

for $0 \leq x \leq \frac{\pi}{2}$.

Marking Rubric for Part 1 (1 Mark)

- 1 mark for correctly computing $C = 2$ and writing the normalized PDF.
- 0 marks if either the constant or the final PDF is wrong.

Part 2: Generating Random Numbers Using Uniform Random Numbers

We use the inverse transform method.

Step 1: Cumulative Distribution Function

$$F(x) = \int_0^x 2 \sin t \cos t dt$$

$$F(x) = \int_0^x \sin(2t) dt$$

$$F(x) = \frac{1 - \cos(2x)}{2}$$

$$F(x) = \sin^2 x$$

Marking Rubric for Step 1 (1.5 Marks)

- 1.5 marks for correctly deriving $F(x) = \sin^2 x$.
- 0.5 marks if the integral setup is correct but simplification contains errors.
- 0 marks for wrong setup and wrong result.

Step 2: Inverse Transformation

Let $u \sim U(0, 1)$. The student may equate the CDF to the uniform variable using either form of the identity:

1. **Direct Form:** $u = \sin^2 x$
2. **Double-Angle Form:** $u = \frac{1 - \cos(2x)}{2}$

Marking Rubric for Step 2 (0.5 Marks)

- 0.5 marks for correctly equating either form of the CDF to the uniform variable u .
- 0 otherwise (Binary marking).

Step 3: Solve for x

Depending on the form chosen in Step 2, the following solutions are equally valid:

- **From $\sin^2 x$:**

$$x = \arcsin(\sqrt{u})$$

- **From double-angle identity:**

$$2u = 1 - \cos(2x) \implies \cos(2x) = 1 - 2u$$

$$x = \frac{1}{2} \arccos(1 - 2u)$$

Both expressions are valid generators for the domain $0 \leq x \leq \frac{\pi}{2}$.

Marking Rubric for Step 3 (0.5 Marks)

- 0.5 marks for the correct final generator formula in any of the above forms.
- 0 otherwise (Binary marking).